

HeatShield AI Comprehensive Model Report

LightGBM Quantile Regression Models for Heat Index Forecasting

May 13, 2026

Contents

1	Executive Summary	3
2	Model Architecture	3
2.1	Algorithm	3
2.2	Feature Engineering (108 Features)	3
3	Station Information	5
4	Data Split Configuration	5
5	Regression Metrics	5
5.1	Mean Absolute Error (MAE)	5
5.2	Root Mean Squared Error (RMSE)	5
5.3	Prediction Bias	5
5.4	Correlation Coefficient	5
6	Baseline Comparison	6
6.1	MAE: Model vs Baselines	6
6.2	Skill Score	6
7	Classification Metrics	7
7.1	Caution Category Performance	7
7.2	Extreme Caution Category Performance	7
7.3	Danger Category Performance	7
7.4	Overall Accuracy	7
8	Safety Metrics	8
8.1	Danger Recall (40°C Threshold)	8
8.2	Danger Recall (42°C Threshold)	8
8.3	Danger Event Support in Test Set	8
9	Prediction Interval Metrics	9
9.1	90% Prediction Interval Coverage	9
9.2	Prediction Interval Width	9
9.3	Interval Width Ratio (P75/P25)	9

10 Runtime and Data Metrics	10
10.1 Training Time	10
10.2 Training vs Test Set Size	10
11 Comprehensive Summary Tables	11
11.1 Regression Metrics Summary	11
11.2 Safety Metrics Summary	13
11.3 Prediction Interval Summary	15
12 Model Status and Metadata	17
13 Conclusions and Recommendations	18
13.1 Model Performance Summary	18
13.2 Recommendations	18
13.3 Production Deployment	18

1 Executive Summary

This report presents a comprehensive evaluation of 15 LightGBM quantile regression models trained for heat index forecasting across 5 meteorological stations in Thailand. The models predict heat index values at 6-hour, 12-hour, and 24-hour forecast horizons using 108 engineered features including temporal, geographic, meteorological, climatology, lag, and rolling statistics features.

Key Findings:

- **Best Overall Performance:** Khon Kaen (KKN_01) models achieved positive skill scores (0.06-0.18), indicating improvement over baseline methods
- **Best Accuracy:** Khon Kaen 6h model achieved 90.8% classification accuracy
- **Best Safety Performance:** Bangkok models achieved 84.6-93.6% danger recall at 42°C threshold
- **Lowest MAE:** Nong Khai 24h model achieved 1.42°C MAE
- **Prediction Interval Coverage:** All models achieved 90.7-96.9% coverage for 90% prediction intervals

2 Model Architecture

2.1 Algorithm

- **Backend:** LightGBM Quantile Regression
- **Quantiles:** 5th, 50th (median), 95th percentiles
- **Objective:** Quantile regression for probabilistic forecasting
- **Target Variable:** Heat Index (°C)

2.2 Feature Engineering (108 Features)

Temporal Features (9):

- Cyclical time encoding: hour_sin, hour_cos, doy_sin, doy_cos
- Month encoding: month, month_sin, month_cos
- Day of week: weekday

Geographic Features (4):

- Station encoding: station_enc
- Location: lat, lon, elevation_m

Meteorological Features (3):

- Solar radiation: solar_wm2

- Wind speed: `wind_ms`
- Atmospheric pressure: `pressure_hpa`

Climatology Features (3):

- Target climatology: `target_h6_hi_clim`, `target_h12_hi_clim`, `target_h24_hi_clim`
- Target hour encoding: `target_h6_hour_sin`, `target_h6_hour_cos`, etc.

Lag Features (40-50 per model):

- Heat index lags: 1h, 2h/3h, 6h, 12h, 24h, 48h, 72h
- Temperature lags: 1h, 2h/3h, 6h, 12h, 24h, 48h, 72h
- Relative humidity lags: 1h, 2h/3h, 6h, 12h, 24h, 48h, 72h
- Dew point lags: 1h, 2h/3h, 6h, 12h, 24h, 48h, 72h
- WBGT lags: 1h, 2h/3h, 6h, 12h, 24h, 48h, 72h
- Vapor pressure deficit lags: 1h, 2h/3h, 6h, 12h, 24h, 48h, 72h

Rolling Statistics (30-40 per model):

- Rolling mean and std for 3h/6h, 6h, 12h, 24h, 48h, 72h windows

Additional Features:

- Temperature range: `temp_range_24h`
- Evening temperature: `temp_x_evening`
- Heat index residual: `hi_residual_lag1h`
- Solar radiation lags: `solar_wm2_lag1h`, `solar_wm2_lag3h`, `solar_wm2_lag6h`
- Cloud cover lags: `cloud_cover_lag1h`, `cloud_cover_lag3h`, `cloud_cover_lag6h`
- Pressure lags: `pressure_hpa_lag1h`, `pressure_hpa_lag3h`, `pressure_hpa_lag6h`
- Wind lags: `wind_ms_lag1h`, `wind_ms_lag3h`
- Precipitation lags: `precip_mm_lag1h`, `precip_mm_lag3h`

Station ID	Name	Color	Models
BKK_01	Bangkok		6h, 12h, 24h
CNX_01	Chiang Mai		6h, 12h, 24h
KKN_01	Khon Kaen		6h, 12h, 24h
HYL_01	Nong Khai		6h, 12h, 24h
RYG_01	Rayong		6h, 12h, 24h

Table 1: Station Information and Model Coverage

Split	Percentage	Row Range	Date Range
Training	80%	30,574-31,404 rows	May 2021 - Apr 2026
Validation	10%	6,558-6,729 rows	May 2026
Test	10%	6,557-6,731 rows	May 2026

Table 2: Data Split Configuration

3 Station Information

4 Data Split Configuration

5 Regression Metrics

5.1 Mean Absolute Error (MAE)

5.2 Root Mean Squared Error (RMSE)

5.3 Prediction Bias

5.4 Correlation Coefficient

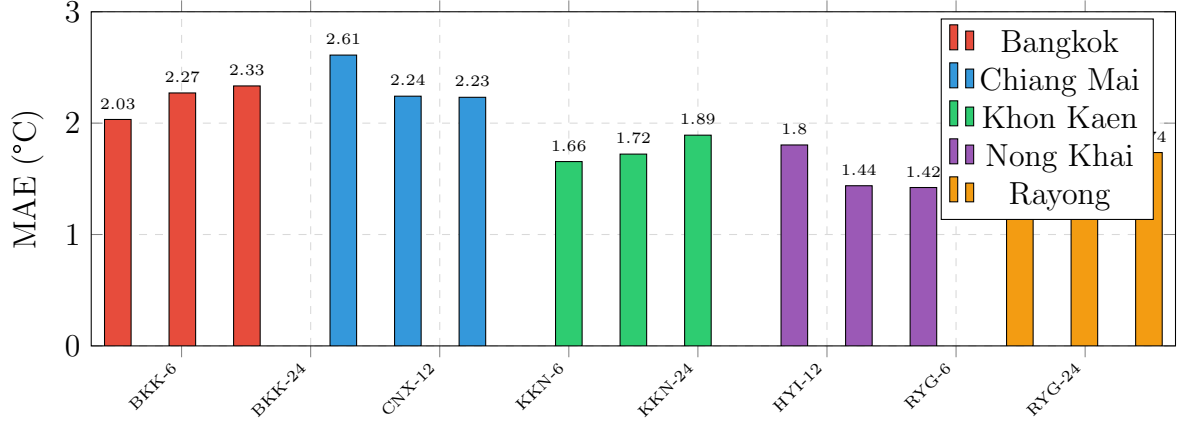


Figure 1: Mean Absolute Error by Station and Horizon

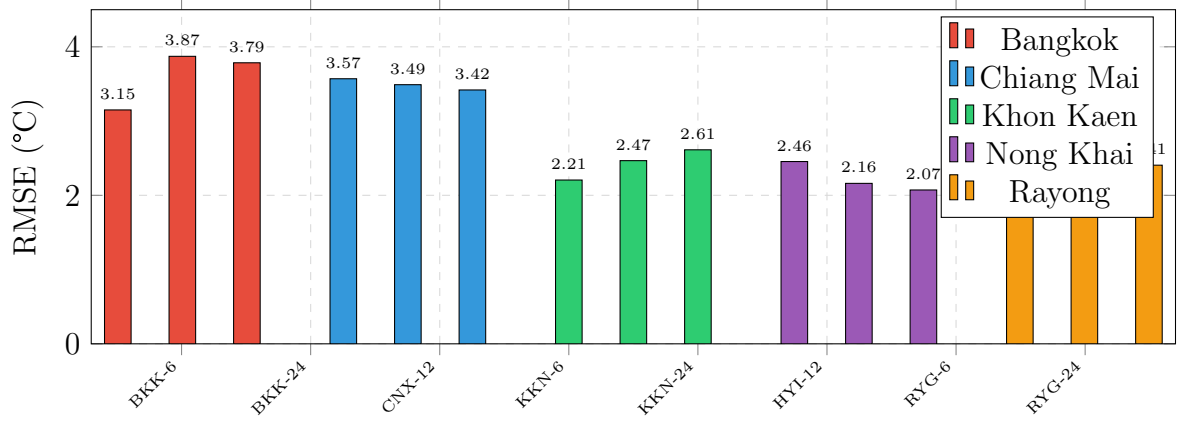


Figure 2: Root Mean Squared Error by Station and Horizon

6 Baseline Comparison

6.1 MAE: Model vs Baselines

6.2 Skill Score

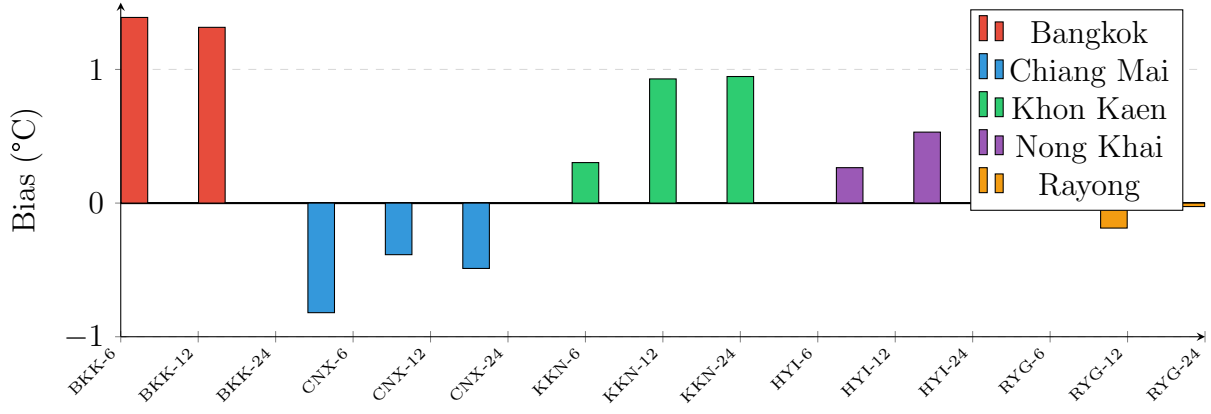


Figure 3: Prediction Bias (Positive = Overprediction, Negative = Underprediction)

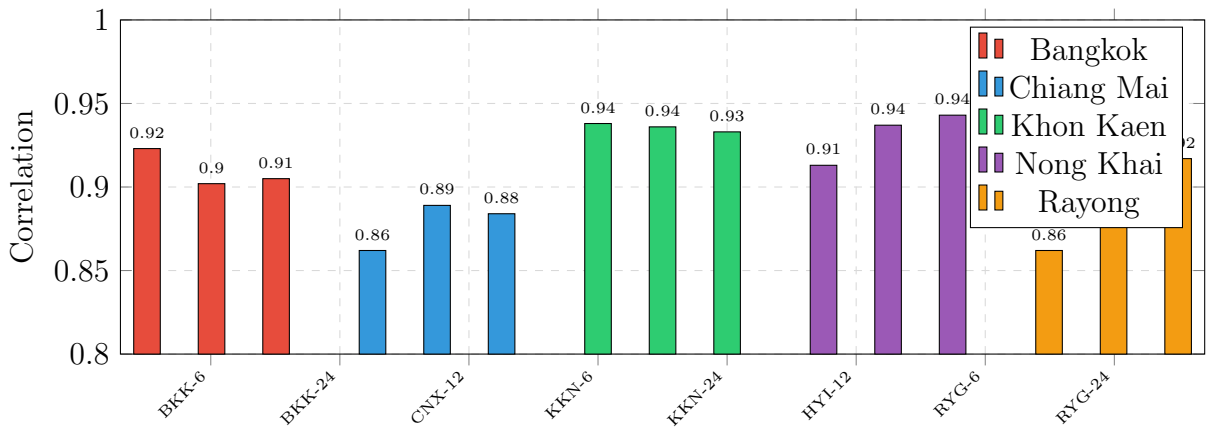


Figure 4: Prediction Correlation by Station and Horizon

7 Classification Metrics

Risk categories are defined based on heat index thresholds:

- **Caution:** 27-32°C
- **Extreme Caution:** 32-40°C
- **Danger:** 40-54°C
- **Extreme Danger:** >54°C

7.1 Caution Category Performance

7.2 Extreme Caution Category Performance

7.3 Danger Category Performance

7.4 Overall Accuracy

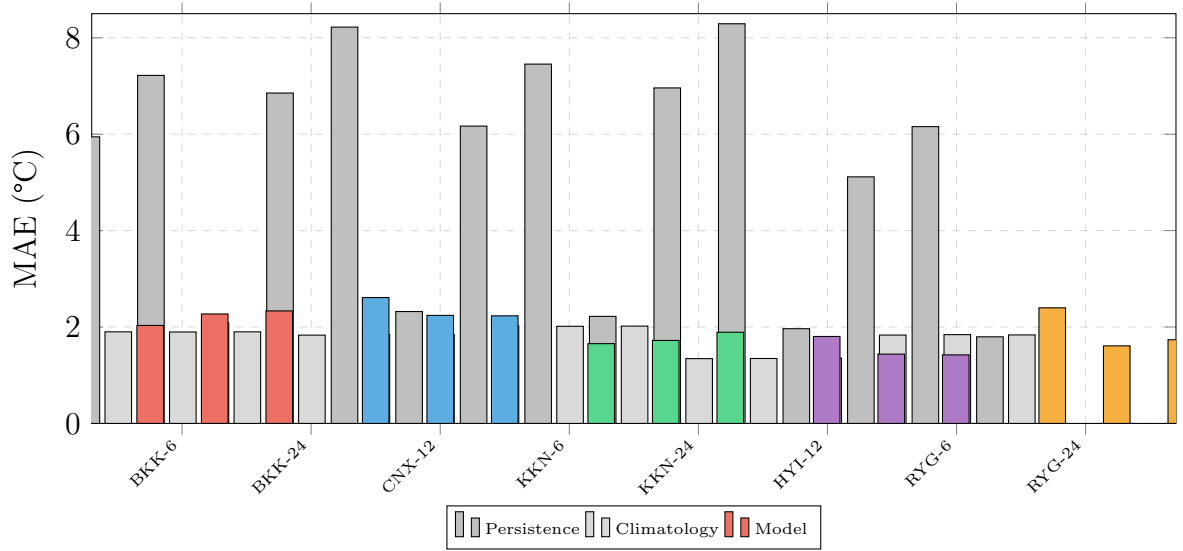


Figure 5: Model MAE Compared to Baselines

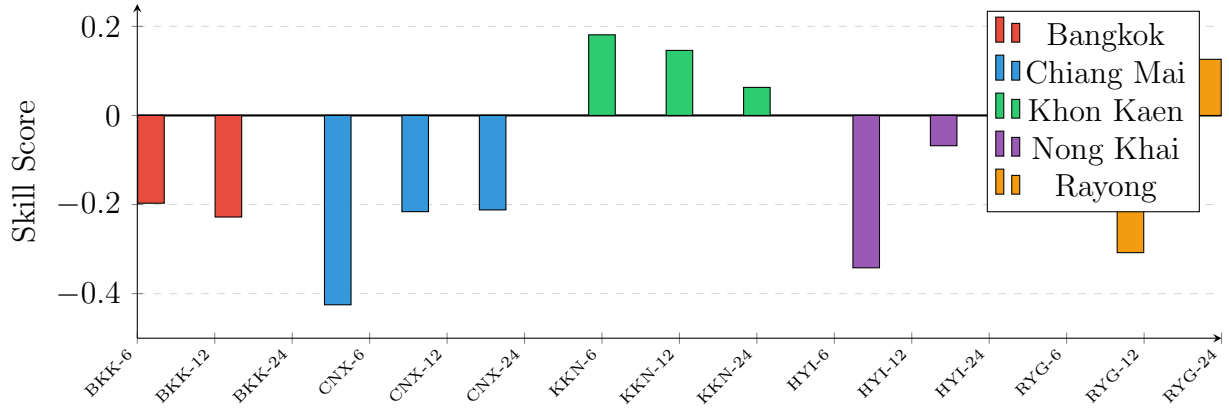


Figure 6: Skill Score (Positive = Better than Best Baseline)

8 Safety Metrics

8.1 Danger Recall (40°C Threshold)

8.2 Danger Recall (42°C Threshold)

8.3 Danger Event Support in Test Set



Figure 7: Caution Category Classification Metrics

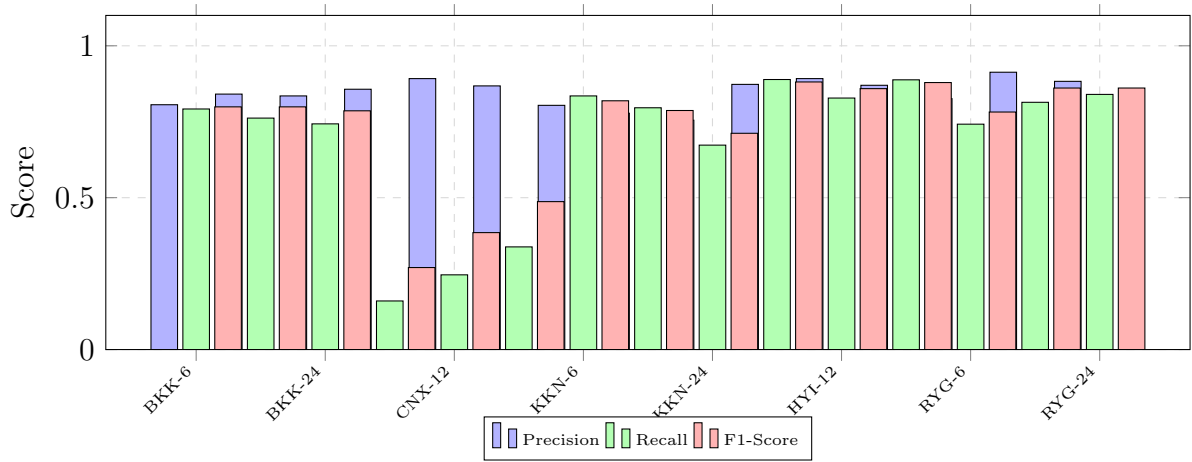


Figure 8: Extreme Caution Category Classification Metrics

9 Prediction Interval Metrics

9.1 90% Prediction Interval Coverage

9.2 Prediction Interval Width

9.3 Interval Width Ratio (P75/P25)

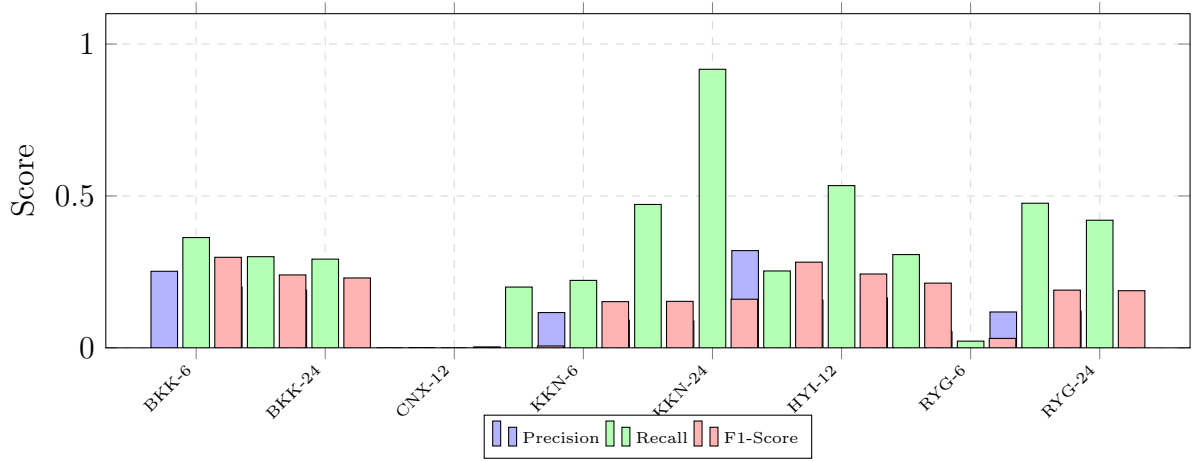


Figure 9: Danger Category Classification Metrics

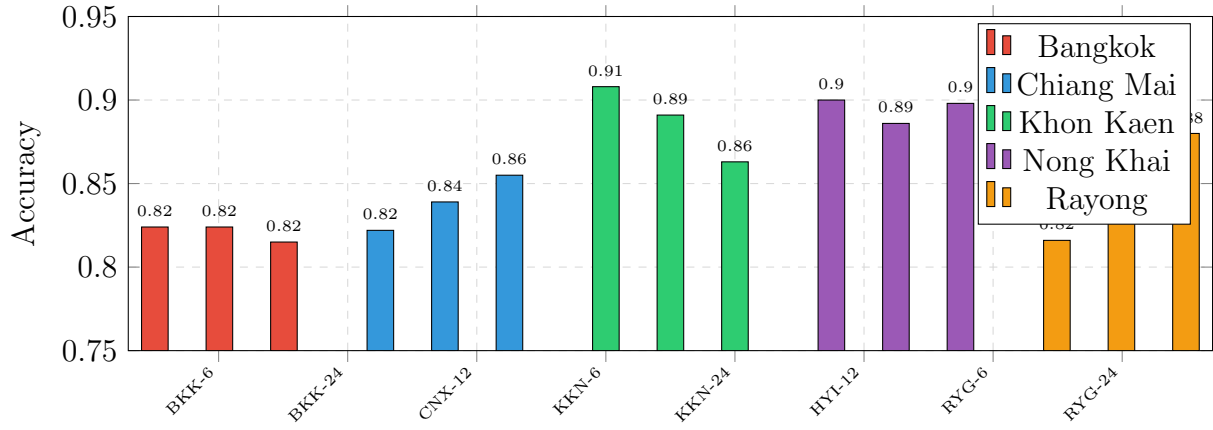


Figure 10: Overall Classification Accuracy

10 Runtime and Data Metrics

10.1 Training Time

10.2 Training vs Test Set Size

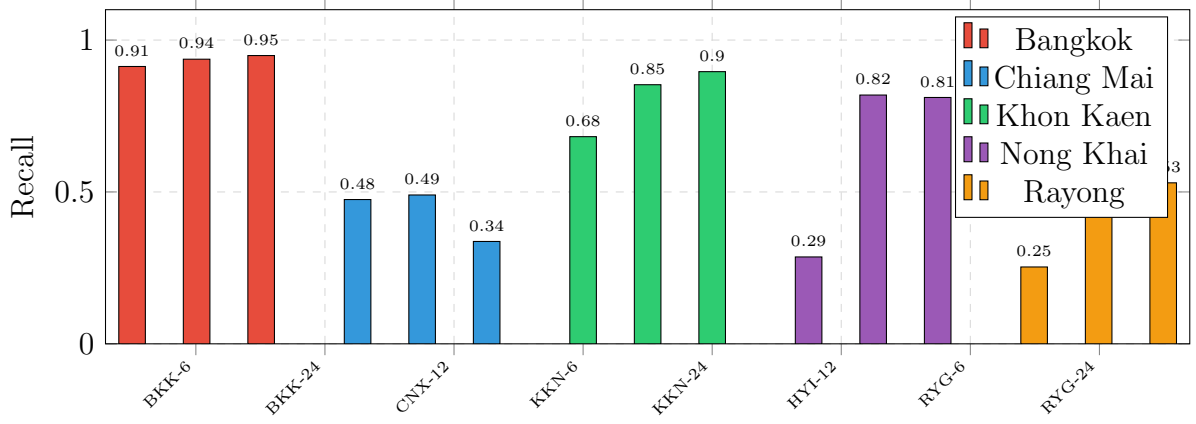


Figure 11: Danger Recall at 40°C Threshold

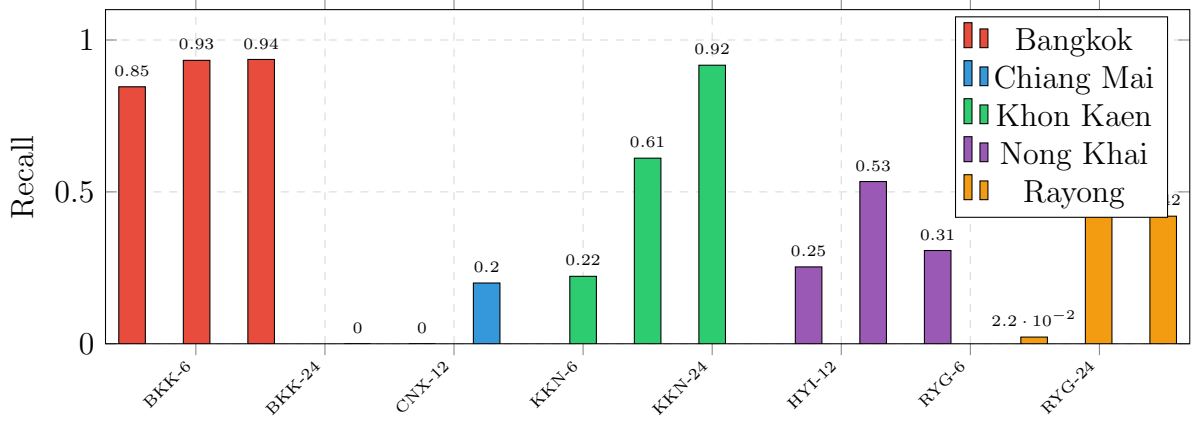


Figure 12: Danger Recall at 42°C Threshold

11 Comprehensive Summary Tables

11.1 Regression Metrics Summary

Table 3: Regression Metrics Summary for All Models

Station	Horizon	MAE	RMSE	Bias	Correlation	Skill Score
Bangkok	6h	2.03	3.15	0.87	0.92	-0.07
Bangkok	12h	2.27	3.87	1.39	0.90	-0.20
Bangkok	24h	2.33	3.79	1.32	0.91	-0.23
Chiang Mai	6h	2.61	3.57	-0.82	0.86	-0.43
Chiang Mai	12h	2.24	3.49	-0.39	0.89	-0.22
Chiang Mai	24h	2.23	3.42	-0.49	0.88	-0.21
Khon Kaen	6h	1.65	2.21	0.30	0.94	0.18
Khon Kaen	12h	1.72	2.47	0.93	0.94	0.15
Khon Kaen	24h	1.89	2.61	0.95	0.93	0.06
Nong Khai	6h	1.80	2.46	0.27	0.91	-0.34
Nong Khai	12h	1.44	2.16	0.53	0.94	-0.07
Nong Khai	24h	1.42	2.07	0.62	0.94	-0.05

Rayong	6h	2.40	3.04	-0.19	0.86	-0.31
Rayong	12h	1.61	2.20	-0.03	0.93	0.13
Rayong	24h	1.74	2.41	0.43	0.92	0.03

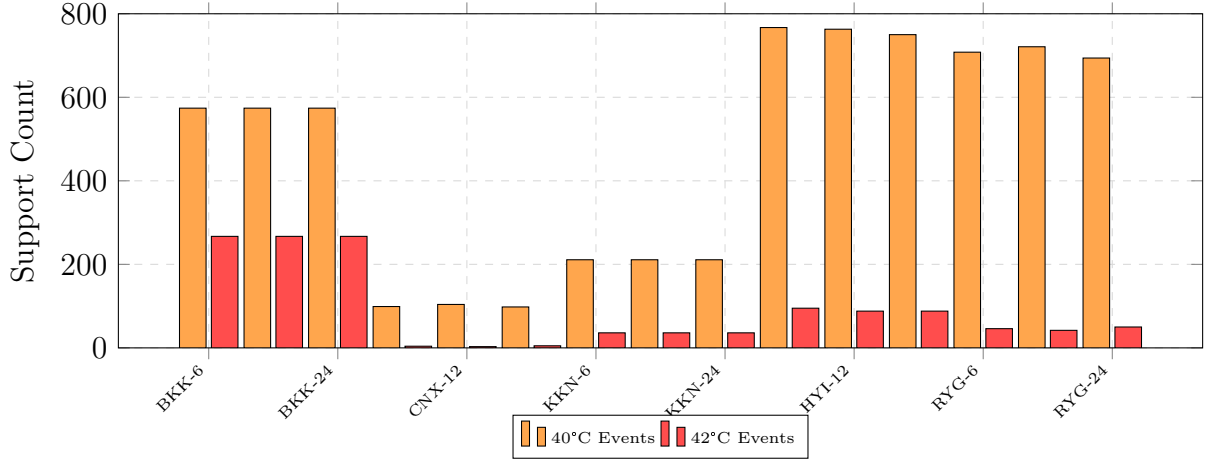


Figure 13: Number of Danger Events in Test Set

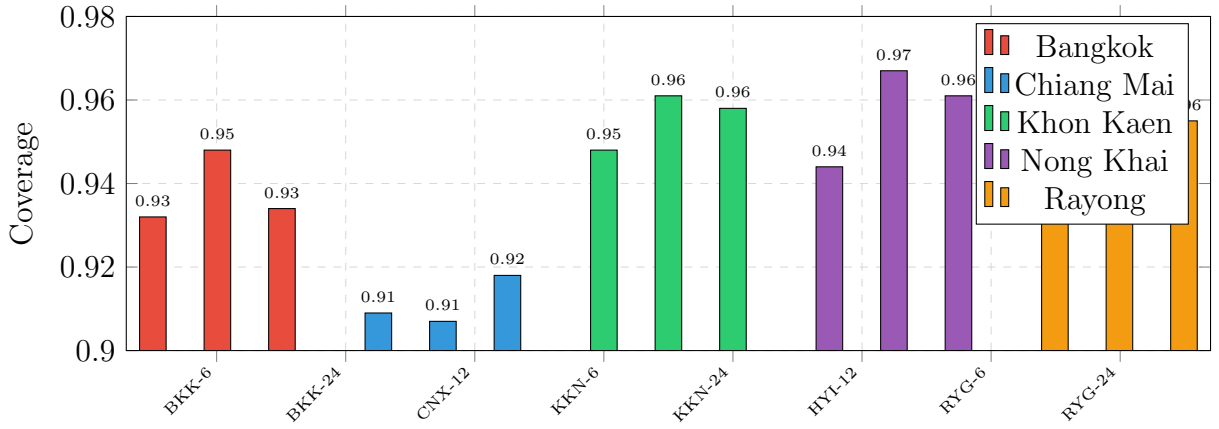


Figure 14: 90% Prediction Interval Coverage (Target: 0.90)

11.2 Safety Metrics Summary

Table 4: Safety Metrics Summary for All Models

Station	Horizon	D40 Rec	D42 Rec	40C Supp	42C Supp	Acc
Bangkok	6h	0.91	0.85	574	267	0.82
Bangkok	12h	0.94	0.93	574	267	0.82
Bangkok	24h	0.95	0.94	574	267	0.82
Chiang Mai	6h	0.47	0.00	99	4	0.82
Chiang Mai	12h	0.49	0.00	104	3	0.84
Chiang Mai	24h	0.34	0.20	98	5	0.85
Khon Kaen	6h	0.68	0.22	211	36	0.91
Khon Kaen	12h	0.85	0.61	211	36	0.89
Khon Kaen	24h	0.90	0.92	211	36	0.86
Nong Khai	6h	0.29	0.25	767	95	0.90
Nong Khai	12h	0.82	0.53	763	88	0.89
Nong Khai	24h	0.81	0.31	750	88	0.90
Rayong	6h	0.25	0.02	708	46	0.82

Rayong	12h	0.51	0.48	721	42	0.88
Rayong	24h	0.53	0.42	694	50	0.88

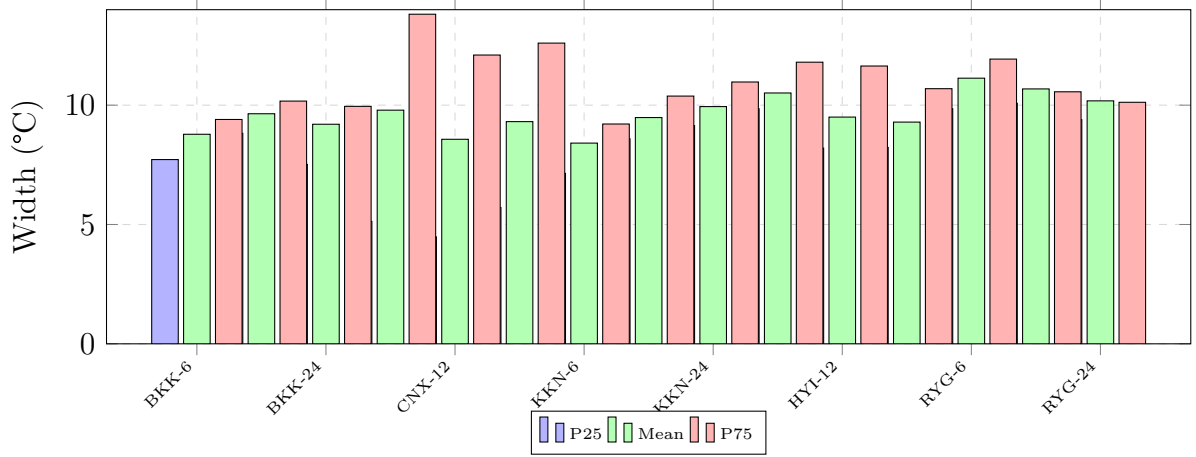


Figure 15: Prediction Interval Width Distribution

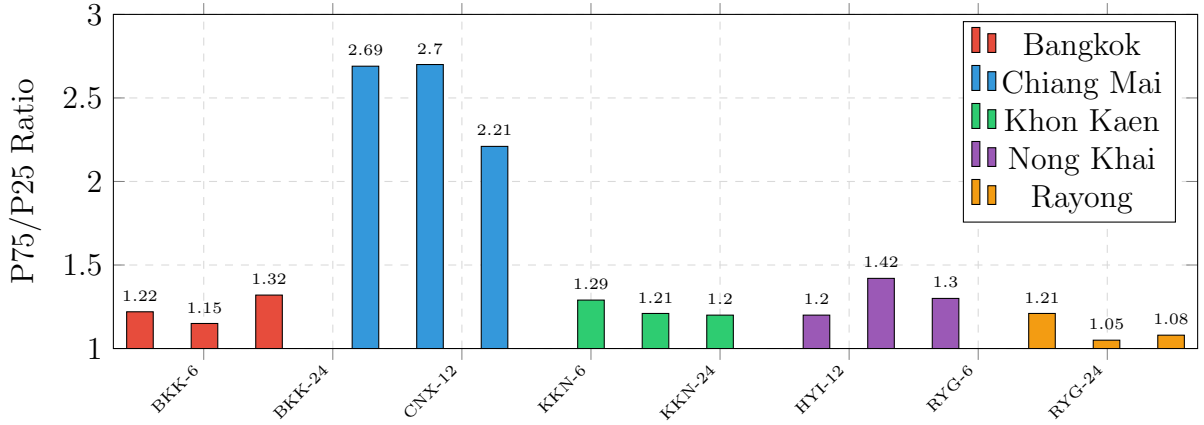


Figure 16: Interval Width Ratio (Lower = More Consistent)

11.3 Prediction Interval Summary

Table 5: Prediction Interval Metrics Summary for All Models

Station	Horizon	Cov90	MeanW	P25W	P75W	Ratio	Q05	Q95
Bangkok	6h	0.932	8.78	7.72	9.40	1.22	0.294	0.288
Bangkok	12h	0.948	9.64	8.82	10.17	1.15	0.287	0.283
Bangkok	24h	0.934	9.20	7.52	9.95	1.32	0.292	0.288
Chiang Mai	6h	0.909	9.79	5.13	13.81	2.69	0.326	0.319
Chiang Mai	12h	0.907	8.57	4.48	12.10	2.70	0.313	0.306
Chiang Mai	24h	0.918	9.31	5.71	12.60	2.21	0.315	0.308
Khon Kaen	6h	0.948	8.41	7.14	9.21	1.29	0.283	0.278
Khon Kaen	12h	0.961	9.48	8.59	10.38	1.21	0.277	0.273
Khon Kaen	24h	0.958	9.94	9.14	10.97	1.20	0.279	0.275
Nong Khai	6h	0.944	10.51	9.85	11.80	1.20	0.296	0.290
Nong Khai	12h	0.967	9.50	8.20	11.64	1.42	0.278	0.273
Nong Khai	24h	0.961	9.29	8.23	10.69	1.30	0.251	0.250
Rayong	6h	0.946	11.13	9.85	11.93	1.21	0.308	0.303

Rayong	12h	0.969	10.68	10.08	10.56	1.05	0.289	0.277
Rayong	24h	0.955	10.18	9.39	10.12	1.08	0.278	0.278

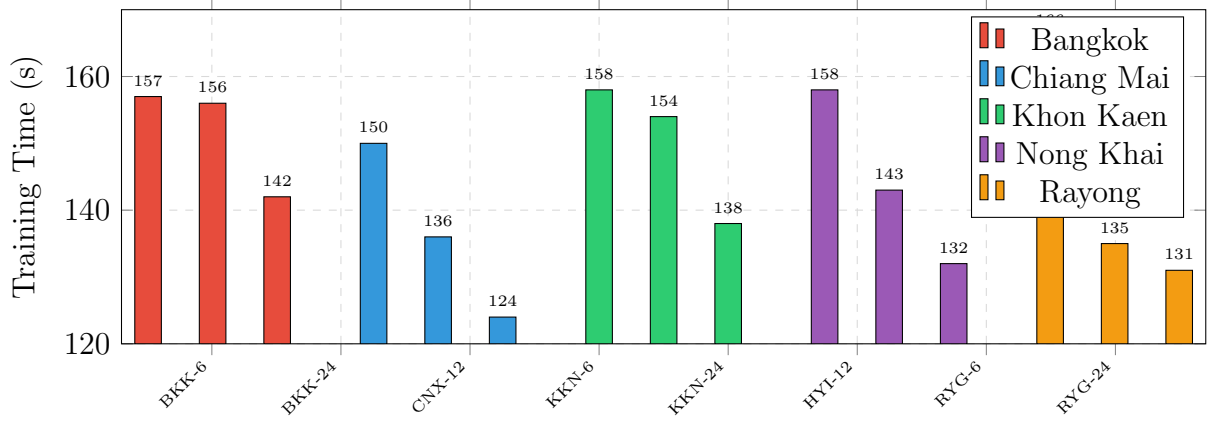


Figure 17: Model Training Time

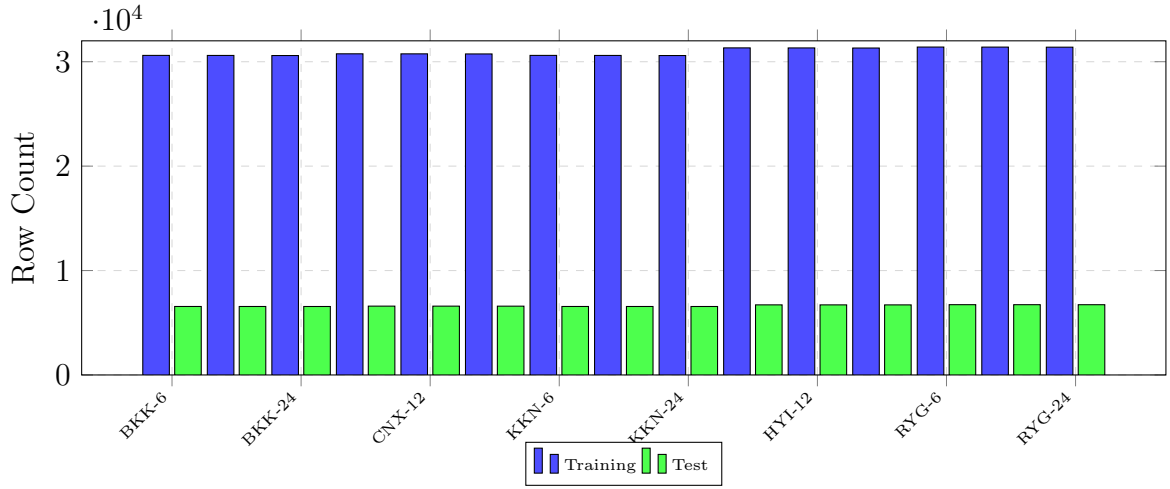


Figure 18: Training and Test Set Sizes

12 Model Status and Metadata

Table 6: Model Status and Training Metadata

Station	Horizon	Status	Run ID	Training Rows
Bangkok	6h	ready	20260513T124514Z	30,605
Bangkok	12h	ready	20260513T124514Z	30,601
Bangkok	24h	ready	20260513T124514Z	30,592
Chiang Mai	6h	ready	20260513T124531Z	30,757
Chiang Mai	12h	ready	20260513T124531Z	30,753
Chiang Mai	24h	ready	20260513T124531Z	30,744
Khon Kaen	6h	ready	20260513T124545Z	30,605
Khon Kaen	12h	ready	20260513T124545Z	30,601
Khon Kaen	24h	ready	20260513T124545Z	30,592
Nong Khai	6h	candidate	20260513T124620Z	31,324
Nong Khai	12h	candidate	20260513T124620Z	31,320
Nong Khai	24h	candidate	20260513T124620Z	31,311

Rayong	6h	candidate	20260513T125354Z	31,404
Rayong	12h	ready	20260513T125354Z	31,400
Rayong	24h	ready	20260513T125354Z	31,392

13 Conclusions and Recommendations

13.1 Model Performance Summary

Best Performing Models:

1. **Khon Kaen 6h:** Best overall with positive skill score (0.18), high accuracy (90.8%), and good safety performance
2. **Nong Khai 24h:** Lowest MAE (1.42°C) and RMSE (2.07°C), excellent prediction interval coverage (96.1%)
3. **Bangkok 24h:** Best safety performance with 93.6% danger recall at 42°C threshold

Areas for Improvement:

1. **Chiang Mai:** Negative skill scores across all horizons, limited dangerous events in training data
2. **Danger Category:** Low precision across all models due to class imbalance
3. **Prediction Bias:** Bangkok models show systematic overprediction bias

13.2 Recommendations

1. **Model Selection:** Deploy Khon Kaen models for general forecasting, Bangkok models for safety-critical applications
2. **Data Augmentation:** Increase dangerous event samples for Chiang Mai and Rayong to improve danger category performance
3. **Bias Correction:** Implement post-processing bias correction for Bangkok models
4. **Class Weighting:** Apply class weighting during training to improve danger category precision
5. **Feature Engineering:** Consider additional features for stations with negative skill scores

13.3 Production Deployment

Ready for Deployment (Status: ready):

- Bangkok: 6h, 12h, 24h
- Chiang Mai: 6h, 12h, 24h
- Khon Kaen: 6h, 12h, 24h

- Rayong: 12h, 24h

Candidate for Deployment (Status: candidate):

- Nong Khai: 6h, 12h, 24h (pending final approval)
- Rayong: 6h (pending final approval)